MockKernel — UI/UX Options (`screens.md`)

> **How to use this file**
>
> 1. Read the 6 **Theme Options** (Section 2) and pick **one**.
> 2. Pick one option from each **Layout Decision** (Section 3) — or accept the defaults.
> 3. Fill in the **Selection Sheet** (Section 4).
> 4. Paste the **Master LLM Prompt** (Section 5) + the chosen theme block + `screen.md` into your LLM.
>
> Every theme is a complete design system (colors, type, shape, motion, component rules), so the whole UI stays consistent. Stack assumed: **React (Vite) + TypeScript + Tailwind CSS + Zustand + Monaco Editor + Socket.io client**.

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1. Quick Comparison

ID	Theme	Mode	Vibe	Best if you want…	Build effort
A	Midnight Terminal	Dark	Hacker / IDE	Devs feel at home, looks like VS Code	Low
B	Clean Studio	Light	Minimal / SaaS	Professional, recruiter-friendly demo	Low
C	Neon Glass	Dark	Futuristic / glassmorphism	"Wow" factor in demos	Medium–High
D	Neo-Brutalist Pop	Light	Bold / playful	Memorable, student-community feel	Medium
E	Arena Esports	Dark	Competitive / high-energy	Contest + leaderboard drama	Medium
F	Warm Paper Focus	Light (warm)	Calm / editorial	Interview practice, low stress	Low
★	Dual Mode (A + B)	Toggle	Pro	Safe default, dark/light switch	Medium

Recommendation for a 15-day sprint with 6 engineers: A (or **★ Dual Mode**) — fewest custom components, Monaco looks native, least design risk.

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2. Theme Options

Each theme defines: palette · verdict colors · typography · shape & spacing · motion · Monaco theme · key-screen treatment · Tailwind tokens · ready-to-paste LLM prompt.

> **Rule across all themes:** verdict states must never rely on color alone — always pair color with an icon + label (`✓ Accepted`, `✗ Wrong Answer`, `⏱ TLE`, `● Pending`).

THEME A — Midnight Terminal 🖥️

Vibe: Developer-native. Feels like a code editor. Dense, fast, keyboard-friendly. **Inspiration:** VS Code, GitHub Dark, LeetCode dark, Warp terminal.

Palette

Token	Hex	Use
`bg-base`	`#0D1117`	App background
`bg-surface`	`#161B22`	Cards, panels
`bg-elevated`	`#21262D`	Modals, dropdowns, hover
`border`	`#30363D`	Dividers
`text-primary`	`#E6EDF3`	Main text
`text-muted`	`#8B949E`	Secondary text
`accent`	`#58A6FF`	Links, primary buttons, focus ring
`accent-alt`	`#BC8CFF`	AI interview highlights
`success`	`#3FB950`	Accepted
`danger`	`#F85149`	Wrong Answer / errors
`warning`	`#D29922`	TLE / rate-limit
`info`	`#79C0FF`	Pending / running
`flag`	`#A371F7`	Plagiarism flagged

Typography

- UI: `Inter`, system-ui, sans-serif
- Code / numbers / verdicts: `JetBrains Mono` or `Fira Code`

- Scale: 12 / 14 / 16 / 20 / 28 px · headings 600 weight

Shape & spacing

- Radius: `6px` (cards) · `4px` (inputs/buttons)
- Spacing: compact (4px base grid), panel padding `12–16px`
- Shadows: none — use 1px borders only
- Icons: `lucide-react`, 16px, stroke 1.5

Motion: fast and subtle (120–150ms). Pending verdict shows a blinking cursor `▌` instead of a spinner.

Monaco theme: `vs-dark` customised with `bg-surface` background, or `github-dark`.

Key screens

- **Arena:** edge-to-edge split panes with draggable gutters, a thin status bar at the bottom (language · cursor line:col · socket status · timer).
- **Console panel:** terminal style — `$ run` prompt, monospaced output, colored stdout/stderr.
- **Interview:** chat looks like a terminal log with speaker prefixes `ai>` and `you>`; mic button is a small pill, not a big orb.
- **Leaderboard:** dense table, rank in mono, top 3 get a thin colored left border.
- **Admin:** sidebar nav + data tables, minimal chrome.

Tailwind tokens

```ts
// tailwind.config.ts → theme.extend
colors: {
  base: '#0D1117', surface: '#161B22', elevated: '#21262D', line: '#30363D',
  ink: '#E6EDF3', muted: '#8B949E', accent: '#58A6FF', ai: '#BC8CFF',
  ok: '#3FB950', bad: '#F85149', warn: '#D29922', run: '#79C0FF', flag: '#A371F7',
},
fontFamily: { sans: ['Inter','system-ui','sans-serif'], mono: ['JetBrains Mono','m
borderRadius: { card: '6px', ctl: '4px' },
```

LLM prompt block

```
THEME: Midnight Terminal (dark, developer-native).
Palette: bg #0D1117, surface #161B22, elevated #21262D, border #30363D, text
#E6EDF3, muted #8B949E,
accent #58A6FF, AI accent #BC8CFF, success #3FB950, danger #F85149, warning
#D29922, pending #79C0FF, flagged #A371F7.
Fonts: Inter for UI, JetBrains Mono for code, verdicts and numbers.
Shape: 6px radius cards, 4px controls, 1px borders, no shadows, compact spacing
(4px grid).
Motion: 120-150ms, blinking-cursor for pending state. Icons: lucide-react 16px.
Monaco: github-dark style. Console panel styled like a terminal. Chat styled
like a terminal log (ai> / you>).
Always pair verdict colors with icon + text label.
```

THEME B — Clean Studio 🌼

Vibe: Bright, airy, professional. Looks like a modern SaaS product. **Inspiration:** Linear (light), Vercel dashboard, Stripe, Notion.

Palette

Token	Hex	Use
`bg-base`	`#FAFAFA`	App background
`bg-surface`	`#FFFFFF`	Cards, panels
`bg-subtle`	`#F4F4F5`	Hover, table stripes
`border`	`#E4E4E7`	Dividers
`text-primary`	`#18181B`	Main text
`text-muted`	`#71717A`	Secondary
`accent`	`#4F46E5`	Primary buttons, links (indigo)
`accent-alt`	`#0EA5E9`	AI interview
`success`	`#16A34A`	Accepted
`danger`	`#DC2626`	Wrong Answer
`warning`	`#D97706`	TLE
`info`	`#2563EB`	Pending
`flag`	`#9333EA`	Plagiarism

Typography

- UI: `Inter` · Code: `Geist Mono` / `JetBrains Mono`
- Scale: 13 / 14 / 16 / 20 / 32 px · generous line-height (1.6)

Shape & spacing

- Radius: `10px` cards, `8px` controls, `999px` badges
- Spacing: comfortable (8px grid), card padding `20–24px`
- Shadows: `0 1px 2px rgba(0,0,0,.05)`, hover `0 4px 12px rgba(0,0,0,.08)`

Motion: 180–220ms ease-out; skeleton shimmer loaders; subtle fade/slide on route change.

Monaco theme: `vs` (light) or `github-light`; editor sits inside a rounded bordered card.

Key screens

- **Arena:** problem statement on the left in a white card, editor right, console as a collapsible bottom drawer.

- **Interview:** friendly chat bubbles (AI left/grey, user right/indigo), large centered mic button with soft pulse while listening.
- **Leaderboard:** clean table with avatar initials and a progress bar for score.
- **Admin:** top nav + card-based dashboard with stat tiles.

Tailwind tokens

```ts
colors: {
  base: '#FAFAFA', surface: '#FFFFFF', subtle: '#F4F4F5', line: '#E4E4E7',
  ink: '#18181B', muted: '#71717A', accent: '#4F46E5', ai: '#0EA5E9',
  ok: '#16A34A', bad: '#DC2626', warn: '#D97706', run: '#2563EB', flag: '#9333EA',
},
borderRadius: { card: '10px', ctl: '8px' },
boxShadow: { card: '0 1px 2px rgba(0,0,0,.05)', lift: '0 4px 12px rgba(0,0,0,.08)'
```

LLM prompt block

```
THEME: Clean Studio (light, minimal SaaS).
Palette: bg #FAFAFA, surface #FFFFFF, subtle #F4F4F5, border #E4E4E7, text
#18181B, muted #71717A,
accent indigo #4F46E5, AI accent sky #0EA5E9, success #16A34A, danger #DC2626,
warning #D97706, pending #2563EB, flagged #9333EA.
Fonts: Inter (UI), Geist Mono / JetBrains Mono (code).
Shape: 10px radius cards, 8px controls, pill badges, soft shadows, comfortable
8px-grid spacing.
Motion: 180-220ms ease-out, skeleton shimmer loaders, subtle fades.
Monaco: light theme inside a rounded bordered card. Chat bubbles: AI grey left,
user indigo right, large pulsing mic button.
Always pair verdict colors with icon + text label.
```

THEME C — Neon Glass 🌌

Vibe: Futuristic, premium, "AI product" look. Gradients, blur, glow. **Inspiration:** Arc browser, Raycast, OpenAI/Anthropic-style dark gradients, Framer templates.

Palette

Token	Hex	Use
`bg-base`	`#07070F`	Background (with radial gradient blobs)
`glass`	`rgba(255,255,255,0.05)`	Panels (with `backdrop-blur-xl`)
`glass-border`	`rgba(255,255,255,0.10)`	Panel borders
`text-primary`	`#F5F5FF`	Main text
`text-muted`	`#9CA3C7`	Secondary
`accent-1`	`#7C3AED`	Violet (primary)
`accent-2`	`#06B6D4`	Cyan (secondary / AI)
`accent-3`	`#F472B6`	Pink highlights
`success`	`#34D399`	Accepted
`danger`	`#FB7185`	Wrong Answer
`warning`	`#FBBF24`	TLE
`info`	`#60A5FA`	Pending
`flag`	`#C084FC`	Plagiarism

Gradients: primary button `linear-gradient(135deg,#7C3AED,#06B6D4)` · background blobs of violet/cyan at 20–30% opacity.

Typography

- Headings: `Space Grotesk` or `Sora` (600–700) · Body: `Inter` · Code: `JetBrains Mono`

Shape & spacing

- Radius: `16px` cards, `12px` controls
- Glass panels: blur 16–24px, 1px translucent border, inner top highlight
- Glow: `box-shadow: 0 0 24px rgba(124,58,237,.35)` on primary actions / active items

Motion: 250–350ms spring easing (Framer Motion), animated gradient on AI "thinking", confetti/glow burst on **Accepted**.

Monaco theme: custom dark with transparent background so the glass panel shows through (`editor.background: #00000000`).

Key screens

- **Arena:** floating glass panels over gradient background; verdict appears as an animated glowing badge.
- **Interview:** central **voice orb** (animated gradient that pulses with audio level), transcript fades in beneath it; manual input slides up as a glass bar.
- **Leaderboard:** top 3 as a podium with glow; the rest as translucent rows; rank changes animate.
- **Admin:** glass sidebar, stat tiles with sparkline charts.

Cautions: needs `backdrop-filter` support, test performance on low-end laptops; keep text contrast ≥ 4.5:1 on glass; offer `prefers-reduced-motion` fallback.

Tailwind tokens

```ts
colors: {
  base: '#07070F', ink: '#F5F5FF', muted: '#9CA3C7',
  violet: '#7C3AED', cyan: '#06B6D4', pink: '#F472B6',
  ok: '#34D399', bad: '#FB7185', warn: '#FBBF24', run: '#60A5FA', flag: '#C084FC',
},
backgroundImage: { brand: 'linear-gradient(135deg,#7C3AED,#06B6D4)' },
borderRadius: { card: '16px', ctl: '12px' },
boxShadow: { glow: '0 0 24px rgba(124,58,237,.35)' },
```

LLM prompt block

```
THEME: Neon Glass (dark, futuristic glassmorphism).
Background #07070F with blurred violet/cyan radial gradient blobs. Panels:
rgba(255,255,255,0.05) with backdrop-blur-xl and 1px rgba(255,255,255,0.10)
border.
Accents: violet #7C3AED, cyan #06B6D4, pink #F472B6; primary buttons use
gradient 135deg violet->cyan with soft glow.
Verdicts: success #34D399, danger #FB7185, warning #FBBF24, pending #60A5FA,
flagged #C084FC.
Fonts: Space Grotesk for headings, Inter body, JetBrains Mono code. Radius 16px
cards / 12px controls.
Motion: Framer Motion springs 250-350ms, glow burst on Accepted, animated
gradient while AI is thinking. Respect prefers-reduced-motion.
Interview screen: central animated voice orb + fading transcript. Leaderboard:
podium for top 3 with glow.
Monaco: transparent-background dark theme. Keep text contrast >= 4.5:1.
```

THEME D — Neo-Brutalist Pop 🎨

Vibe: Bold, playful, loud. Thick borders, hard shadows, flat bright colors. Matches the hand-drawn feel of your architecture diagram. **Inspiration:** Gumroad, Figma community pages, Excalidraw, Neobrutalism.dev.

Palette

Token	Hex	Use
`bg-base`	`#FFF8E7`	Cream background
`surface`	`#FFFFFF`	Cards
`ink`	`#111111`	Text AND all borders
`accent-yellow`	`#FFD60A`	Primary actions
`accent-pink`	`#FF6B9D`	Highlights
`accent-blue`	`#4D96FF`	AI interview / links
`accent-mint`	`#6BCB77`	Success
`danger`	`#FF4D4D`	Wrong Answer
`warning`	`#FF9F1C`	TLE
`flag`	`#B388FF`	Plagiarism

Typography

- Headings: `Space Grotesk` / `Archivo Black` (700–900) · Body: `DM Sans` · Code: `JetBrains Mono`

Shape & spacing

- Border: **2–3px solid** `#111` on everything
- Radius: `8px` (slightly rounded) or `0` (pure brutal)
- Shadow: **hard offset** `4px 4px 0 #111`; pressed state `0 0 0` + translate(4px,4px)
- Each section gets a different pastel/bright fill

Motion: snappy 100ms; buttons "press down" into their shadow; verdict badge "stamps" in with a small rotate + scale.

Monaco theme: `vs` light with a cream background, wrapped in a thick-bordered card with hard shadow.

Key screens

- **Arena:** each panel is a colored card with a thick border; Run/Submit are big chunky buttons (yellow / mint).
- **Interview:** chat bubbles as sticker-like boxes with hard shadows; mic is a big yellow button with a thick outline.
- **Leaderboard:** each row a bordered strip; #1 gets a yellow fill and crown emoji.
- **Admin:** bold stat tiles in different colors, chunky tables.

Cautions: can feel noisy if every element has a different color — limit each screen to 2–3 fills.

Tailwind tokens

```ts
colors: {
  cream: '#FFF8E7', ink: '#111111', yellow: '#FFD60A', pink: '#FF6B9D',
  blue: '#4D96FF', mint: '#6BCB77', bad: '#FF4D4D', warn: '#FF9F1C', flag: '#B388F
},
boxShadow: { hard: '4px 4px 0 #111', 'hard-sm': '2px 2px 0 #111' },
borderWidth: { 3: '3px' },
borderRadius: { card: '8px' },
```

LLM prompt block

```
THEME: Neo-Brutalist Pop (light, bold, playful).
Background cream #FFF8E7, surfaces white, ALL text and borders #111111 with 2-
3px solid borders.
Hard offset shadows (4px 4px 0 #111), pressed buttons translate into their
shadow. Radius 8px.
Fills: yellow #FFD60A (primary), pink #FF6B9D, blue #4D96FF (AI), mint #6BCB77
(success), red #FF4D4D (wrong), orange #FF9F1C (TLE), lilac #B388FF (flagged).
Max 2-3 fill colors per screen.
Fonts: Space Grotesk/Archivo Black headings, DM Sans body, JetBrains Mono code.
Motion: 100ms snappy; verdict badge stamps in with slight rotation. Chat bubbles
are sticker-style bordered boxes.
Always pair verdict colors with icon + text label.
```

THEME E — Arena Esports 🏆

Vibe: High-energy competition. Scoreboards, ranks, tension. Makes contests feel like a live event.
Inspiration: Codeforces/AtCoder dashboards reimagined, Valorant/League UI, F1 timing screens.

Palette

Token	Hex	Use
`bg-base`	`#0B0E14`	Background
`surface`	`#131822`	Panels
`surface-2`	`#1B2230`	Raised
`border`	`#2A3345`	Dividers
`text-primary`	`#EAF0FF`	Main
`muted`	`#8A96B2`	Secondary
`accent`	`#FF3D5A`	Primary (hot red)
`gold`	`#FFC53D`	Rank #1 / highlights
`silver`	`#C0C8D8`	Rank #2
`bronze`	`#CD7F32`	Rank #3
`success`	`#22D3A0`	Accepted
`danger`	`#FF5C7A`	Wrong Answer
`warning`	`#FFA94D`	TLE
`info`	`#4DA3FF`	Pending
`flag`	`#B07CFF`	Plagiarism

Typography

- Headings/numbers: `Rajdhani` or `Orbitron` (600–700, uppercase, wide tracking) · Body: `Inter` · Code: `JetBrains Mono`
- Big tabular numerals for timer and scores

Shape & spacing

- Radius: `4px` with **angled corners** (`clip-path` on key cards/buttons)
- Accent lines: 2px colored top border on panels; skewed section headers
- Shadow: subtle inner glow on active elements

Motion: 150–200ms; leaderboard rows **slide to new positions** on rank change (FLIP animation); score tick-up counter; timer turns red + pulses in last 5 minutes; toast banner for "First Blood" (first

solver of a problem).

Monaco theme: `vs-dark` with `bg-surface`; thin red active-line highlight.

Key screens

- **Arena:** top HUD with big countdown, problem tabs `A B C D` with solved/attempted dots, collapsible mini-leaderboard on the right.
- **Leaderboard:** hero screen — podium for top 3 (gold/silver/bronze), live movement arrows ▲▼, per-problem cells showing ✓ + time or − attempts.
- **Interview:** more subdued — same dark palette, red accent replaced by calm blue/teal for a less aggressive feel.
- **Admin Live Monitor:** looks like a broadcast control room (live feed, queue depth, flagged alerts).

Cautions: keep the Interview section calmer; avoid overly aggressive red on error states that could stress candidates.

Tailwind tokens

```ts
colors: {
  base: '#0B0E14', surface: '#131822', raised: '#1B2230', line: '#2A3345',
  ink: '#EAF0FF', muted: '#8A96B2', accent: '#FF3D5A',
  gold: '#FFC53D', silver: '#C0C8D8', bronze: '#CD7F32',
  ok: '#22D3A0', bad: '#FF5C7A', warn: '#FFA94D', run: '#4DA3FF', flag: '#B07CFF',
},
fontFamily: { display: ['Rajdhani','sans-serif'], sans: ['Inter','sans-serif'], mo
```

LLM prompt block

```
THEME: Arena Esports (dark, competitive, high-energy).
Palette: bg #0B0E14, surface #131822, raised #1B2230, border #2A3345, text
#EAF0FF, muted #8A96B2,
accent hot red #FF3D5A, gold #FFC53D, silver #C0C8D8, bronze #CD7F32, success
#22D3A0, danger #FF5C7A, warning #FFA94D, pending #4DA3FF, flagged #B07CFF.
Fonts: Rajdhani (uppercase, wide tracking) for headings/numbers, Inter body,
JetBrains Mono code. Tabular numerals for timer and scores.
Shape: 4px radius with angled-corner clip-path on key cards; 2px colored top
border on panels.
Motion: FLIP row animations on leaderboard rank change, score count-up, timer
pulses red in the last 5 minutes, "First Blood" banner.
Arena screen: top HUD with big countdown, problem tabs A/B/C with status dots,
collapsible mini leaderboard.
Leaderboard: podium for top 3, rank-movement arrows, per-problem result cells.
Interview screen uses a calmer blue/teal accent instead of red.
```

THEME F — Warm Paper Focus 📖

Vibe: Calm, human, low-stress. Feels like a notebook. Ideal for interview practice and reading problem statements. **Inspiration:** Notion, Medium, Readwise, Anthropic's warm neutrals, Obsidian light.

Palette

Token	Hex	Use
`bg-base`	`#F7F3EC`	Warm paper background
`surface`	`#FFFDF8`	Cards
`subtle`	`#EFE9DD`	Hover / secondary
`border`	`#E0D8C8`	Dividers
`text-primary`	`#2B2A27`	Main
`muted`	`#7A7468`	Secondary
`accent`	`#C2410C`	Terracotta primary
`accent-alt`	`#3F7D6B`	Sage green (AI interview)
`success`	`#3F7D4E`	Accepted
`danger`	`#B23A3A`	Wrong Answer
`warning`	`#B7791F`	TLE
`info`	`#3B6EA5`	Pending
`flag`	`#7C5CA8`	Plagiarism

Typography

- Headings & problem statements: `Source Serif 4` / `Lora` (serif, 600) · UI: `Inter` · Code: `IBM Plex Mono`
- Reading width capped at ~70ch; line-height 1.7

Shape & spacing

- Radius: `12px` cards, `8px` controls
- Spacing: airy (8px grid, large margins)
- Shadows: very soft `0 1px 3px rgba(60,50,30,.08)`

Motion: slow and gentle (250ms); no flashing; typewriter-style streaming for AI replies.

Monaco theme: `solarized-light` or custom warm light (`bg #FFFDF8`).

Key screens

- **Arena:** serif problem statement with generous spacing; editor in a paper-like card; verdict shown as a calm inline note rather than a loud badge.
- **Interview:** the hero screen — large serif question text, soft sage mic button, transcript as a notebook page; "Take your time" micro-copy.
- **Leaderboard:** quiet, list-like; ranks in serif numerals.
- **Admin:** simple, card-based, lots of white space.

Cautions: less "contest energy" — good for practice/interview, weaker for live competition drama.

Tailwind tokens

```ts
colors: {
  paper: '#F7F3EC', surface: '#FFFDF8', subtle: '#EFE9DD', line: '#E0D8C8',
  ink: '#2B2A27', muted: '#7A7468', accent: '#C2410C', sage: '#3F7D6B',
  ok: '#3F7D4E', bad: '#B23A3A', warn: '#B7791F', run: '#3B6EA5', flag: '#7C5CA8',
},
fontFamily: { serif: ['Source Serif 4','serif'], sans: ['Inter','sans-serif'], mon
```

LLM prompt block

```
THEME: Warm Paper Focus (light warm neutrals, calm and editorial).
Palette: bg #F7F3EC, surface #FFFDF8, subtle #EFE9DD, border #E0D8C8, text
#2B2A27, muted #7A7468,
accent terracotta #C2410C, AI accent sage #3F7D6B, success #3F7D4E, danger
#B23A3A, warning #B7791F, pending #3B6EA5, flagged #7C5CA8.
Fonts: Source Serif 4 for headings and problem statements, Inter for UI, IBM
Plex Mono for code. Max reading width 70ch, line-height 1.7.
Shape: 12px radius cards, 8px controls, very soft shadows, airy spacing.
Motion: gentle 250ms, no flashing, typewriter-style streaming for AI replies.
Monaco: warm light theme. Verdict shown as calm inline note. Interview screen is
the hero: large serif question, soft sage mic button, "Take your time" micro-
copy.
Always pair verdict colors with icon + text label.
```

★ THEME G — Dual Mode (Midnight Terminal + Clean Studio)

Idea: Ship **A (dark)** and **B (light)** together using CSS variables + Tailwind `dark:` class, with a toggle in the top bar (default: follow system).

Implementation notes

- Define colors as CSS variables: `--bg`, `--surface`, `--border`, `--ink`, `--muted`, `--accent`, `--ok`, `--bad`, `--warn`, `--run`, `--flag`.
- Tailwind: `colors: { surface: 'rgb(var(--surface) / <alpha-value>)' }` and `darkMode: 'class'`.
- Monaco: switch between `vs-dark` and `vs` via `monaco.editor.setTheme()` on toggle.
- Persist the choice in `uiStore` (Zustand) + `localStorage`.

LLM prompt block

```
THEME: Dual Mode. Implement BOTH "Midnight Terminal" (dark) and "Clean Studio"
(light) using CSS variables
(--bg, --surface, --border, --ink, --muted, --accent, --ok, --bad, --warn, --
run, --flag) and Tailwind darkMode:'class'.
Add a theme toggle in the top bar (default: system preference), persist in
Zustand + localStorage, and switch Monaco between vs-dark and vs.
Use the palettes defined in Theme A and Theme B. Shared components must read
only from CSS variables, never hard-coded hex.
```

---

3. Layout & UX Decisions (pick one per row)

These are independent of the visual theme. Defaults are marked **(default)**.

3.1 Navigation

Option	Description	Good for
N1 — Top bar (default)	Horizontal nav, user menu right	Student side, simple
N2 — Left sidebar	Collapsible icon + label rail	Admin side, many sections
N3 — Hybrid	Top bar for students, sidebar for admin	Most apps *(recommended)*

3.2 Contest Arena / Problem Workspace Layout

Option	Layout	Notes
L1 — Vertical split (default)	Problem left · Editor top-right · Console bottom-right	LeetCode/Codeforces-like, familiar
L2 — Horizontal split	Problem top · Editor + Console below	Better on small laptop heights
L3 — Tabbed single pane	Tabs: Problem / Code / Console / Leaderboard	Best for mobile/tablet
L4 — Focus mode	Editor full-screen, problem in slide-over drawer	For concentration; toggled with `F`

Extras to decide: ☐ draggable resizable panes · ☐ collapsible console · ☐ keyboard shortcuts (`Ctrl+Enter` run, `Ctrl+Shift+Enter` submit) · ☐ font-size control · ☐ vim/emacs keybindings.

3.3 Verdict Panel Style

Option	Description
V1 — Terminal log (default)	Streamed monospaced lines, status header
V2 — Test-case cards	Each test case a card: ✓ / ✗, time, memory, diff on fail
V3 — Banner + details	Big verdict banner on top, expandable details below

3.4 AI Interview Screen

Option	Description	Pairs well with
I1 — Chat thread (default)	Standard chat, mic button beside the input	A, B, D
I2 — Voice orb hero	Large animated orb center, transcript beneath	C
I3 — Split: question card + answer area	One question at a time, big card, answer recorder below	F
I4 — Video-call style	AI avatar tile + user tile, live captions	Any (most work)

Extras to decide: ☐ live transcript while speaking · ☐ text-to-speech for AI replies · ☐ progress indicator (Q3/8) · ☐ "Skip / Repeat question" buttons · ☐ visible fallback banner when Web Speech is

unavailable (E1).

3.5 Leaderboard

Option	Description	Pairs well with
LB1 — Simple table (default)	Rank · user · score · solved	A, B, F
LB2 — ICPC-style grid	Per-problem columns with ✓ time / – attempts	A, E
LB3 — Podium + list	Top-3 podium, rest as list	C, D, E

Extras: ☐ live rank-change animation · ☐ highlight my row · ☐ pin me to bottom if off-screen · ☐ "First Blood" badge.

3.6 Dashboard (Student)

Option	Description
D1 — Stat tiles + lists (default)	KPIs on top, recent activity below
D2 — Activity-first feed	Timeline of submissions/interviews/contests
D3 — Action hub	Big cards: "Solve", "Join Contest", "Start Interview"

3.7 Admin Layout

Option	Description
AD1 — Table-centric (default)	Dense CRUD tables, side drawers for editing
AD2 — Card-centric	Visual cards per contest/problem
AD3 — Control room	Live monitor first, CRUD secondary

3.8 Responsiveness

Option	Description
R1 — Desktop-first (default)	Optimised ≥1024px; editor screens show "use a larger screen" below 768px
R2 — Fully responsive	Tabbed layouts (L3) on mobile, all screens usable

3.9 Feedback & Micro-interactions

☐ Toasts (top-right) · ☐ Skeleton loaders · ☐ Confetti on first Accepted · ☐ Sound effects (opt-in) · ☐ Rate-limit cooldown ring on Run button (E4) · ☐ Socket status dot · ☐ Optimistic UI on submit.

---

4. Selection Sheet (fill this in, then paste with the prompt)

```
PROJECT: MockKernel
THEME:              [ A | B | C | D | E | F | G(Dual) ]
NAVIGATION:         [ N1 | N2 | N3 ]
ARENA LAYOUT:       [ L1 | L2 | L3 | L4 ]
VERDICT PANEL:      [ V1 | V2 | V3 ]
AI INTERVIEW:       [ I1 | I2 | I3 | I4 ]
LEADERBOARD:        [ LB1 | LB2 | LB3 ]
STUDENT DASHBOARD:  [ D1 | D2 | D3 ]
ADMIN LAYOUT:       [ AD1 | AD2 | AD3 ]
RESPONSIVE:         [ R1 | R2 ]
EXTRAS (tick):      [ ] resizable panes  [ ] shortcuts  [ ] TTS  [ ] live
transcript
                    [ ] confetti  [ ] sound  [ ] rank animations  [ ] first
blood
```

Suggested Combos

Combo	Theme	Choices
Safe & Fast	A or G	N3 · L1 · V1 · I1 · LB2 · D1 · AD1 · R1
Demo Showstopper	C	N3 · L1 · V3 · I2 · LB3 · D3 · AD3 · R1
Contest Hype	E	N3 · L1 · V2 · I1 · LB2 + LB3 · D2 · AD3 · R1
Interview-First	F	N1 · L4 · V3 · I3 · LB1 · D3 · AD2 · R2
Friendly Student Club	D	N1 · L1 · V3 · I1 · LB3 · D3 · AD2 · R1
Recruiter-Ready	B	N3 · L1 · V2 · I1 · LB1 · D1 · AD1 · R2

---

5. Master LLM Prompt Template

Paste this, then append: (1) the chosen **theme prompt block** from Section 2, (2) your filled **Selection Sheet**, (3) the contents of `screen.md`.

```
You are a senior frontend engineer and product designer. Build the complete UI
for "MockKernel",
an interactive coding platform with async code execution, adaptive AI voice
interviews, live contests,
plagiarism detection and a real-time leaderboard.

TECH STACK (fixed):
- React 18 + Vite + TypeScript (strict) + Tailwind CSS
- Zustand for state, React Router for routing, Socket.io-client for realtime
- @monaco-editor/react for the editor, Web Speech API for voice, lucide-react
for icons
- Optional: Framer Motion (only if the chosen theme calls for it)

DESIGN SYSTEM:
[PASTE THEME PROMPT BLOCK HERE]

LAYOUT CHOICES:
[PASTE SELECTION SHEET HERE]

SCREENS TO BUILD:
[PASTE screen.md HERE]

STRICT REQUIREMENTS:
1. Put ALL design tokens (colors, fonts, radius, shadows) in tailwind.config.ts
/ CSS variables. No hard-coded hex inside components.
2. Build a reusable component library first: Button, Input, Card, Badge,
VerdictBadge, Modal, Toast, DataTable,
   Skeleton, EmptyState, Timer, Tabs, Drawer, ResizablePanels.
3. Every screen must implement loading, empty, error and success states.
4. Verdict states (PENDING, ACCEPTED, WRONG_ANSWER, TLE, RUNTIME_ERROR,
COMPILE_ERROR, PLAGIARISM_FLAGGED)
   must use color + icon + text label. Never color alone.
5. Code execution UI: POST submission -> receive 202 + submissionId -> show
PENDING -> update via Socket.io
   event `submission:update`; if the socket drops, automatically fall back to
polling GET /api/submissions/:id.
6. Rate limiting (E4): on HTTP 429 show a toast and disable Run/Submit with a
visible cooldown.
7. AI interview (E1/E3): if Web Speech API is unsupported or mic permission is
denied, automatically switch to a
   manual textarea with a visible notice; on Groq 429 show a "AI is busy,
```

```
retrying" state.
8. Accessibility: WCAG AA contrast, visible focus rings, full keyboard
navigation, aria-labels on icon buttons,
   aria-live region for verdict updates, respect prefers-reduced-motion.
9. Use mock data + a typed API layer (src/api/*.ts) with TypeScript interfaces
so the backend can be swapped in later.
10. Folder structure:
src/{api,components,features,pages,store,hooks,lib,styles,types}.

DELIVERABLES (in order):
1. tailwind.config.ts + global CSS (tokens, fonts)
2. Shared component library
3. App shell, routing, route guards (student/admin)
4. Screens in this priority: Auth -> Contest Arena/Problem Workspace -> AI
Interview -> Leaderboard
   -> Student Dashboard -> Admin (Problems, Contests, Live Monitor, Plagiarism)
5. Short README explaining how to change the theme.

Work screen by screen. After each screen, list the files created and wait for
confirmation before continuing.
```

---

6. Shared UX Rules (apply to every theme)

Area	Rule
Verdict feedback	Show status within 100ms of click (optimistic PENDING); final verdict within 10s per PRD
Editor	Persist code per problem in memory/draft; warn before leaving with unsaved code
Rate limit (E4)	Cooldown ring on Run button; message "Slow down — try again in Ns"
Timeout (E2)	TLE message explains "Your code exceeded the 10s limit — check for infinite loops"
Mic fallback (E1)	Inline banner, not a blocking modal; manual input is immediately usable
AI retry (E3)	Typing indicator + "AI is busy, retrying…"; never lose the user's answer
Plagiarism	Students see a neutral "Under review" status; only admins see similarity scores
Realtime status	Always show socket state (connected / reconnecting / polling)
Empty states	Illustration/icon + one-line explanation + primary action
Destructive actions	Confirm modal for delete problem / cancel contest / disqualify
Contrast	Text ⩾ 4.5:1, large text/icons ⩾ 3:1 in both light and dark themes
Motion	Honor `prefers-reduced-motion`; no auto-flashing > 3Hz

7. Per-Screen Treatment Cheat Sheet

Screen	A Terminal	B Studio	C Neon Glass	D Brutalist	E Arena	F Paper
Login/Register	Centered card, mono labels	Split screen with product shot	Glass card on gradient	Chunky bordered form	Dark card with red accent bar	Simple serif form
Dashboard	Dense tiles	Stat cards + charts	Glass tiles + sparklines	Colorful tiles	Rank + upcoming-contest hero	Calm list cards
Problem List	Compact table	Clean table + filters	Translucent rows	Bordered strips	Table with difficulty chips	Reading-list style
Arena	Split panes + status bar	Card panes + drawer console	Floating glass panels	Colored bordered cards	HUD + problem tabs	Serif statement + paper editor
Verdict	Terminal log	Banner + details	Glow badge	Stamped badge	Test-case cards	Inline calm note
Interview	Terminal chat	Friendly bubbles	Voice orb	Sticker bubbles	Calm teal variant	Notebook page
Leaderboard	Dense table	Table w/ progress bars	Podium + glass rows	Bordered strips + crown	ICPC grid + podium	Quiet list
Admin	Sidebar + tables	Top nav + cards	Glass sidebar	Bold tiles	Control-room monitor	Spacious cards

---

8. Decision Checklist Before Handing to the LLM

- [ ] One theme chosen (no mixing palettes, except Dual Mode).
- [ ] Layout options chosen for Arena, Interview and Leaderboard (these three define the product).
- [ ] Desktop-first vs responsive decided.

- [ ] Fonts available (Google Fonts links or `@fontsource` packages) — confirm licensing/offline needs.
- [ ] Monaco theme decided and registered via `monaco.editor.defineTheme`.
- [ ] Extras list finalised (TTS, confetti, sound) — these add scope.
- [ ] `screen.md` attached so the LLM knows every screen and route.
- [ ] Team agrees on API contracts (payload shapes for submissions, interviews, leaderboard) so mock data matches the real backend.

Tip for iterating with the LLM

Ask for **one screen at a time**, review it visually, then say: *"Keep the same theme tokens and components, now build the next screen."* If the look drifts, re-paste the theme block. To try another theme later, only the tokens and the theme block should change, not the components.